

MARS Railway Safety System

Practical Hardware, Software Integration, and Model Training Plan

Item	Detail
Purpose	Feasible and cost-conscious plan for collecting field data and operating MARS end to end.
Prepared for	MARS project implementation and demonstration
Current software state	FastAPI inference service, React dashboard, regional simulator, sensor simulation script, and Hugging Face Space demo
Date	June 14, 2026

Recommended approach

Start with a low-cost pilot on non-critical demonstration track or lab rig, validate sensor quality and model behavior, then harden enclosures, communications, and operating procedures before any field deployment near live railway operations.

1. Executive Summary

MARS is now a region-aware multimodal railway safety system. The software can ingest track sensor telemetry, weather conditions, and video/security signals; apply Indian Railway zone and division context; and return risk severity, recommended action, and explainable feature contributions.

The most practical implementation is a phased architecture: inexpensive sensor nodes collect vibration and temperature; a local edge gateway validates and buffers data; optional camera edge AI handles trespasser or obstruction events; and the MARS API performs risk inference and publishes alerts to dashboards or downstream control systems.

2. What Has Been Implemented in MARS

Area	Completed work
Regional profiles	Added zone profiles for NR, CR, WR, ER, SR, NFR, and NWR with divisions, hazards, and operating factors.
Backend scoring	Rule-based track and weather models now adjust for regional speed, vibration, maintenance, and weather sensitivity.
API contract	Segment metadata supports optional division; predictions include region, division, and the resolved regional profile.
Discovery endpoint	Added GET /regions so clients can list supported railway zones and operating profiles.
Sensor simulator	scripts/simulate_sensors.py now generates region-aware track and weather events and calls both inference endpoints.
Frontend	React dashboard and evaluator include zone/division controls, regional context, and corrected explanation rendering.
Hugging Face Space	Updated to a Gradio multimodal simulator with track, weather, video/security, and region controls.
Tests	Added unit tests for region aliases and region-driven rule-based score differences.

3. End-to-End Working Architecture

The production architecture should stay modular so low-cost pilots and hardened field deployments use the same software contract.

1. Track-side sensor node samples vibration, temperature, optional strain/displacement, GPS/time, and local health data.
2. Node publishes compact JSON or MQTT messages to an edge gateway over wired Ethernet, RS-485, Wi-Fi, LoRaWAN, or cellular depending on the site.
3. Edge gateway validates schema, timestamps data, stores a local buffer, and forwards windows to the MARS inference API.
4. Camera/video subsystem emits detection metadata such as trespasser confidence, obstruction confidence, and camera visibility instead of streaming every frame to the model API.
5. MARS API computes track risk, weather-fusion risk, and security anomaly risk, then publishes alerts and explanations to dashboard, event bus, or maintenance workflow.
6. Operator reviews severity and recommended action; confirmed outcomes are saved for retraining and threshold tuning.

4. Practical Hardware Stack

The following stack is feasible for a college, lab, or low-budget pilot while preserving a path to industrial-grade deployment.

Subsystem	Cost-effective option	Role in MARS
Vibration sensing	Budget: MPU-6050/ADXL345 for demo rig. Better pilot: ADXL355 low-noise 3-axis accelerometer. Industrial: IEPE accelerometer with signal conditioner.	Detect vertical/lateral vibration, impacts, poor support, loosened fittings, and abnormal wheel-track interaction.
Track temperature	Budget/pilot: waterproof DS18B20 digital probe. Industrial: RTD/PT100 with transmitter.	Detect heat-related expansion risk and local thermal stress.
Environmental sensing	Budget: BME280 for temperature, humidity, pressure. Pilot: tipping-bucket rain gauge plus visibility proxy. Industrial: compact weather station with rainfall, wind, visibility.	Feed weather-track fusion: rainfall, visibility, wind, ambient temperature, fog/heat/flood flags.
Video/security	Budget: USB or IP camera on Raspberry Pi. Pilot: PoE IP camera. Industrial: rugged thermal/network camera for low-light or fog-prone sites.	Detect trespassers, track obstruction, platform crowding, and camera visibility issues.
Sensor node MCU	ESP32 for low-cost Wi-Fi/Bluetooth prototypes; STM32/industrial MCU for hardened nodes.	Samples sensors, filters noise, packages messages, and provides local watchdog/health reporting.
Edge gateway	Raspberry Pi 5 for track/weather gateway. Jetson Orin Nano when local video AI is needed.	Buffers data, performs lightweight preprocessing, runs camera detection, and forwards to API.
Connectivity	Lab: Ethernet/Wi-Fi. Sparse field: LoRaWAN for low-rate sensors. Remote high-bandwidth: 4G/5G router.	Keeps sensor data flowing while controlling cost and bandwidth.
Power	Lab mains adapter. Field: solar panel + charge controller + LiFePO4 battery + surge protection.	Supports remote operation and safe shutdown.
Enclosure/mounting	IP65/IP67 enclosure, cable glands, DIN rail, vibration isolation, tamper seal.	Protects electronics and makes maintenance repeatable.

5. Pilot Bill of Materials Tiers

Tier	Hardware set	When to use
Lab demo kit	ESP32, MPU-6050/ADXL345, DS18B20, BME280, USB/IP camera, Raspberry Pi 5, bench power supply.	Lowest cost; proves software flow, UI, data schema, and model behavior. Not suitable for harsh field use.
Practical field pilot	ADXL355, DS18B20 or RTD module, rain gauge, BME280/weather module, PoE IP camera, Raspberry Pi 5 gateway, 4G router, IP65 enclosure, surge protection.	Good balance of data quality and cost. Recommended first real pilot on controlled site.
Hardened deployment	Industrial accelerometer, RTD/PT100, certified weather station, rugged PoE/thermal camera, Jetson edge AI, industrial gateway, UPS/solar, IP67 enclosure.	Higher cost; use after pilot validates value and operating procedure.

Cost discipline

Do not start with industrial hardware everywhere. Begin with inexpensive sensors on a lab rig to validate sampling rate, feature extraction, API integration, and dashboard behavior. Upgrade only the sensors whose data quality directly affects risk decisions.

6. Data Acquisition Plan

Data type	Recommended collection rate	Features used by MARS
Vibration	100-1000 Hz at node; send 1-10 second windows or aggregate features	Mean, standard deviation, peak, RMS, dominant frequency, lateral/vertical ratio
Speed	Per train pass or 1 Hz during simulation; source from train telemetry, GPS, axle counter, or manual feed in pilot	Mean, max, speed ratio against permitted speed
Track temperature	0.1-1 Hz	Mean, max, heat flag, deviation from regional baseline
Weather	1-5 minute interval	Rainfall, visibility, ambient temperature, wind speed, flood/fog/heat flags
Video/security	Camera stream locally; API receives detections only	Trespasser confidence, obstruction confidence, crowding confidence, camera visibility
Node health	30-60 second interval	Battery, RSSI, packet loss, uptime, enclosure temperature

7. Software Integration With Current MARS Code

The current software already has the right seams for hardware integration. Hardware should not call model internals directly; it should publish normalized events that match the existing schemas.

Layer	Responsibility	Current integration point
Sensor node firmware	Reads sensor values, applies calibration, timestamps packets, sends JSON/MQTT.	Maps to SensorEvent fields: speed, acceleration, vibration_vertical, vibration_lateral, track_temperature.
Weather station adapter	Normalizes station output and local alerts.	Maps to WeatherEvent fields: rainfall_mm, visibility_m, temperature_c, wind_speed_kmph, hazard_flags.
Camera edge AI	Runs object detection locally and sends annotations or summarized confidence values.	Maps to VideoEvent/SecurityAnomalyRequest or Hugging Face demo security inputs.
Edge gateway	Batches events into windows, validates data, retries failed uploads, stores local buffer.	Calls /track-risk/predict, /weather-risk/predict, /security-anomaly/predict, and /regions.
MARS backend	Feature engineering, model inference, regional adjustment, explanation, event publishing.	src/mars_ml_pipeline/services/app.py and feature/model modules.
Frontend/HF demo	Operator-facing controls and visualization.	React dashboard for local demo; Hugging Face Space for public multimodal demonstration.

8. Example Field Message Contracts

Keep messages compact, explicit, and versioned. The gateway can translate protocol-specific payloads into these application contracts.

Track sensor event example:

```
{ "timestamp": "2026-06-14T08:00:00Z", "segment_id": "CR-SEG-001", "train_id": "IR-10001", "speed": 96, "acceleration": 0.2, "vibration_vertical": 0.42, "vibration_lateral": 0.30, "track_temperature": 37 }
```

Segment metadata example:

```
{ "segment_id": "CR-SEG-001", "line_id": "CR-MUMBAI", "region": "CR", "division": "Mumbai", "asset_type": "track", "age_years": 22, "maintenance_score": 0.68, "curvature_degree": 3.2, "max_permitted_speed": 105 }
```

Weather event example:

```
{ "region": "CR", "segment_id": "CR-SEG-001", "rainfall_mm": 80, "visibility_m": 900, "temperature_c": 34, "wind_speed_kmph": 70, "hazard_flags": [ "flood" ] }
```

9. Model Training Method

The training workflow should combine synthetic bootstrapping, controlled pilot data, and operator-confirmed labels. Synthetic data is useful for initial demos; field labels are essential before operational use.

1. Define the prediction target: normal, caution, high_risk. Tie labels to inspection outcomes, speed restrictions, maintenance findings, and verified incidents.
2. Generate baseline data using `src/mars_ml_pipeline/data/generate_synthetic.py` for early model behavior and CI tests.
3. Collect pilot data from sensor nodes and weather/video adapters. Store raw events and derived features separately.
4. Build track features with speed statistics, acceleration magnitude, vertical/lateral vibration statistics, temperature mean, segment age, maintenance score, curvature, and permitted speed.
5. Build weather-fusion features by adding rainfall, visibility, ambient temperature, wind speed, and hazard flags.
6. Train models with the existing scripts: `train_track_risk.py` and `train_weather_risk.py`. Export artifacts with version metadata.
7. Evaluate using confusion matrix, precision/recall for high-risk class, false-alarm rate, missed-event rate, and region-wise performance slices.

8. Deploy artifacts through MARS_TRACK_MODEL_PATH and MARS_WEATHER_MODEL_PATH. If absent, the deterministic regional rule model remains the fallback.
9. Continuously retrain with confirmed field outcomes, but keep a model registry and rollback path.

Model	Training data	Training route	Deployment artifact
Track model	track_sensor_events.csv	train_track_risk.py	mars-track-risk.joblib
Weather fusion model	track_sensor_events.csv + weather_events.csv	train_weather_risk.py	mars-weather-risk.joblib
Security detector	Video annotations or camera detections	AnnotationSecurityDetector now; future YOLO/edge detector export	security detector artifact or rules

10. Region-Wise Practical Behavior

Zone group	Typical condition	MARS adjustment
CR	Ghat sections, suburban density, monsoon disruption	Increase vibration/weather sensitivity and reduce speed tolerance slightly.
NFR	Hills, rivers, landslides, remote maintenance	Highest weather and vibration sensitivity; prioritize rain, landslide, and visibility alerts.
WR/NWR	Heat, dust, long routes, higher-speed trunk sections	Watch thermal stress and dust visibility; speed tolerance can be higher where permitted.
ER/SR	Waterlogging, coastal humidity, bridge approaches, cyclone exposure	Increase weather sensitivity and corrosion/maintenance weighting.
NR	Fog-prone plains and dense traffic	Increase low-visibility/fog and congestion awareness.

11. Deployment Phases

Phase	Duration	Work	Safety boundary
Phase 0: lab proof	1-2 weeks	Build sensor rig, run simulator, validate API and dashboard, verify data schema.	No field safety reliance.
Phase 1: controlled pilot	4-8 weeks	Install 2-5 nodes on non-critical or supervised section; compare sensor readings with manual inspection.	Daily calibration checks; no automatic operational action.
Phase 2: extended pilot	2-3 months	Add weather and video/security; tune thresholds by region; collect labels for retraining.	Human-in-the-loop advisories only.
Phase 3: hardened rollout	After pilot acceptance	Industrial enclosures, rugged sensors, redundancy, formal maintenance SOP.	Integrate with operations only after safety review.

12. Calibration, Reliability, and Maintenance

- Calibrate accelerometers on installation and after any physical maintenance.

- Record mounting position, orientation, rail/fastener condition, and sensor serial number for every segment.
- Use gateway buffering so outages do not destroy data continuity.
- Flag stale sensors, low battery, weak signal, high packet loss, and impossible values as data-quality alerts.
- Keep manual inspection feedback in the dataset; otherwise the model will learn sensor noise but not real maintenance outcomes.
- Use role-based access, TLS, signed firmware, and per-device credentials before field deployment.

13. Acceptance Checklist

Area	Acceptance criterion	Status
Hardware	Sensor node survives vibration/heat/rain test; enclosure has strain relief and surge protection.	Pass / Fail
Data quality	Timestamps are synchronized; missing data and impossible values are detected.	Pass / Fail
API integration	Gateway can call /track-risk/predict, /weather-risk/predict, /security-anomaly/predict, and /regions.	Pass / Fail
Dashboard	Operators can see region/division, severity, score, and recommended action.	Pass / Fail
Model	High-risk recall and false-alarm rate meet pilot targets; region slices reviewed.	Pass / Fail
Operations	Human review process exists before any speed restriction or field action.	Pass / Fail

14. Recommended Minimum Viable Pilot

Pilot recipe

Use Raspberry Pi 5 as the gateway, ESP32-based sensor nodes for low-rate telemetry, ADXL355 for vibration on the primary pilot node, DS18B20 for track temperature, BME280 plus rain gauge for weather, one PoE/IP camera for security metadata, and the existing MARS FastAPI service plus React dashboard. This is practical, cost-effective, and directly maps to the current codebase.

Minimum working software commands:

- Run backend: `uvicorn mars_ml_pipeline.services.app:create_app --factory --reload --port 8000`
- Run regional simulator: `python scripts/simulate_sensors.py --base-url http://localhost:8000`
- Run frontend: `npm run dev` from frontend/
- Train track model: `python -m mars_ml_pipeline.training.train_track_risk --input data/generated/track_sensor_events.csv --artifact-dir artifacts/track-risk/v0.1.0`
- Train weather model: `python -m mars_ml_pipeline.training.train_weather_risk --sensor-input data/generated/track_sensor_events.csv --weather-input data/generated/weather_events.csv --artifact-dir artifacts/weather-risk/v0.1.0`

Appendix A: Hardware Reference Sources

Reference	Source
Raspberry Pi 5	Official Raspberry Pi product page: https://www.raspberrypi.com/products/raspberry-pi-5/
NVIDIA Jetson Orin Nano	Official NVIDIA developer kit specifications: https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit/
ESP32	Espressif ESP32 product page: https://www.espressif.com/en/products/socs/esp32
RAK LoRaWAN gateway	RAK WisGate Edge Lite 2 product page: https://www.rakwireless.com/en-us/products/lpwan-gateways-and-concentrators/rak7268-wisgate-edge-lite-2
ADXL355 accelerometer	Analog Devices product/datasheet page: https://www.analog.com/en/products/adxl355.html
BME280 environmental sensor	Bosch Sensortec datasheet: https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bme280-ds002.pdf
DS18B20 temperature probe	Analog Devices/Maxim datasheet: https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf
Axis thermal/network cameras	Axis camera product categories: https://www.axis.com/products/network-cameras
Axis Perimeter Defender	Axis edge-based intrusion analytics: https://www.axis.com/products/axis-perimeter-defender

Appendix B: Practical Notes

- Exact hardware brands can change during procurement. Select equivalent parts that meet the same sensing, enclosure, power, and communication requirements.
- For live railway environments, installation must follow railway safety approvals, electrical isolation requirements, right-of-way rules, and maintenance authority procedures.
- The MARS system should provide decision support first. Automated operational control should only be considered after formal validation, fail-safe review, and regulator/operator approval.